

Upcoming Webinar

Surprises in time series analysis

DATE

Monday, October 26, 2026

TIME

6:00 PM - 7:30 PM ET



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Abstract

I will present a statistical framework for identifying anomalies in three time series settings: unusual observations within a single historical series, unusual series within a large collection, and real-time surveillance of incoming data streams.

In each case, anomalies are identified using a probabilistic approach based on "surprisal values" - equal to minus the log (conditional) density of each observation - with extreme value theory used to model the tail of the surprisal distribution and avoid strong parametric assumptions.

The methods will be illustrated using the [{weird}](#) package for R, with examples including French and US mortality rates, and pharmaceutical sales.

Software

The methods will be illustrated using the [weird](#) package for R.

References

- Hyndman, Rob J. 2026. That's Weird: Anomaly Detection Using R. <https://OTexts.com/weird>.
- Hyndman, Rob J, and David T Frazier. 2026. "Anomaly Detection Using Surprisals." <http://robjhyndman.com/publications/surprisals.html>.